Joseph Paul Koyi

Software Engineer | Backend & Systems

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Profile

Software engineer with hands-on experience building backend systems, developer tooling, and production web applications, focused on solving complex problems programmatically and shipping production-grade software.

Work Experience

Nov 2024–May 2025 **Software Engineering Intern**, *Ericsson*, Budapest, Hungary

- Developed custom Java static-analysis detectors with SpotBugs to enforce SEI CERT secure-coding rules, achieving over 99% true-positive rate in evaluation.
- Tested detectors against large open-source Java codebases to improve precision and reduce false positives and regressions.
- Implemented detector registration and XML configuration, and contributed through GitHub/GitLab pull-request workflows.

Projects

QualEvents – Event Lifecycle & Operations Platform

- Built an event platform with Next.js, TypeScript, Prisma, and PostgreSQL for contributions and pledges, payment tracking, guest lists, personalized invitations, RSVP, and venue check-in.
- Automated the contributor-to-guest flow using configurable qualification rules, pledge/payment tracking, and auditable review and correction history.
- Integrated Beem SMS and Meta WhatsApp for invitations, reminders, and delivery tracking; added personalized cards, QR/manual-code check-in, and GitHub Actions CI/CD with PostgreSQL integration, smoke/load checks, isolated pre-production, and gated Vercel deployment.

NebulaPower – Solar Financing & Monitoring Platform

- Built a multi-tenant solar financing and monitoring platform with Java 21, Spring Boot, Next.js, and PostgreSQL that lets solar companies manage customers, installations, payment plans, installment payments, and live energy usage from connected solar systems.
- Connected field devices through MQTT to send energy readings, heartbeats, and device status while receiving remote control commands. Delivered real-time dashboards to separate admin and customer applications via WebSocket, with automated service restriction/restoration when payment or security conditions require it.

Fuzzy Parser – Schema-Driven Parsing Engine

- Built a Rust engine that turns messy text, CSV, and Excel files into structured records. Applications define their expected fields once; the engine detects candidate values such as names, phone numbers, emails, dates, amounts, and custom categories, then assigns them using field types, labels, headers, constraints, and surrounding input context.
- Designed it to flag missing, conflicting, or uncertain values instead of silently guessing, while keeping the original source linked to every parsed result. Exposed the same engine through a Rust API, command-line tool, and Node.js/WebAssembly package.

Skills

Languages	Java, Rust, TypeScript, C++, Python, C, SQL
Backend & Data	Spring Boot, Spring Security, REST APIs, JPA/Hibernate, PostgreSQL, Prisma, WebSocket, MQTT
Engineering	Git, Docker, GitHub Actions, Linux, JUnit, Vitest, CI/CD, SDLC, Software Architecture

Education

Sept 2023–Feb 2026 **BSc Computer Science**, *Eötvös Loránd University*, Budapest, Hungary

Certifications

Docker Foundations – Docker Inc. Ubuntu Linux – Canonical